Example

What is the constant term in the expansion of $(x + \frac{1}{x^3})^{12}$?

Solution:

We need the Binomial expansion term $\binom{12}{a}x^a(\frac{1}{x^3})^{12-a}$ to produce a constant. We rewrite as:

$$\binom{12}{a}x^a(x^{-3})^{12-a} = \binom{12}{a}x^a x^{-3(12-a)} = \binom{12}{a}x^{a-3(12-a)}$$

For the power of x to be 0 in the constant term it must:

$$a - 3(12 - a) = 0 \Rightarrow 4a = 36 \Rightarrow a = 9$$

Therefore, the constant is $\binom{12}{9} = 220$.